

Wireless DOORBELL

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Installing a doorbell in an existing building is a difficult task as it involves wiring that can look shabby if not concealed properly. Presented here is a circuit that does not require external wiring and can be placed in small enclosures. This circuit can be used as a call bell in office, too.

Circuit and working

This circuit consists of transmitter and receiver units.

Transmitter: Circuit diagram of the transmitter is shown in Fig. 1. The transmitter circuit is built around 5V voltage regulator 7805 (IC1), encoder HT12E (IC2), a DIP switch (DIP1) and a few other components. IC2 converts 12-bit (8-bit address and 4-bit data) parallel data to serial data, which is available at its DOUT pin.

DIP1 is used to set the address bit either high or low. All 4-bit data pins (AD8 through AD11) are connected to

ground to reduce power consumption because 433MHz RF transmitter module (TX1) uses on-off key (OOK) modulation. When the doorbell pushbutton switch (S2) is pressed, data along with the address is sent serially through wireless transmitter module TX1.

OOK modulation is the binary form of amplitude modulation. When data being sent is low, the transmitter is fully off, suppressing the carrier. In this state TX1 consumes very low current of about 1mA.

When data being sent is high, the transmitter is fully on. In this state current consumption of TX1 is high of about 11mA with 3V power supply.

Receiver: Circuit diagram of the receiver is shown in Fig. 2. The receiver circuit is built around 5V voltage regulator 7805 (IC3), decoder HT12D (IC4), NE555 timer (IC5), melody generator UM66 (IC6), audio amplifier LM386 (IC7) and a few other components.

Serial data transmitted through TX1 is received by RF receiver module RX1. It is fed to pin 14 of the

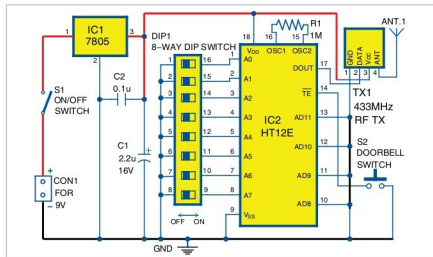


Fig. 1: Circuit diagram of transmitter unit for the wireless doorbell

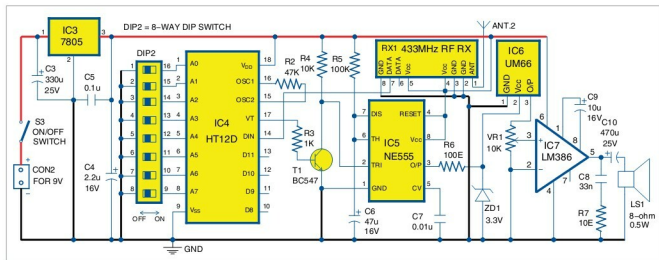


Fig. 2: Circuit diagram of receiver unit for the wireless doorbell